

NUCLEAR PHYSICS

LESSON-13

NUCLEAR PHYSICS

SUB TOPICS

- ⇒ Binding Energy
- ⇒ Nuclear Fission/Fusion
- ⇒ Nuclear Characteristics
- ⇒ Einstein's Mass Energy Equivalence.

Concepts &

KEY POINTS...

Radius of Nucleus $R = R_0 A^{1/3}$

where,

$A \rightarrow$ Mass Number

$$R_0 = 1.1 \text{ fermi}$$

$$R_0 = 1.1 \times 10^{-15} \text{ m}$$

Mass of Proton = Mass of Neutron

- Density of all Nucleus is Same ...

$$\text{Density of Nucleus} \approx 10^{16} \text{ Kg/m}^3$$

Binding Energy:

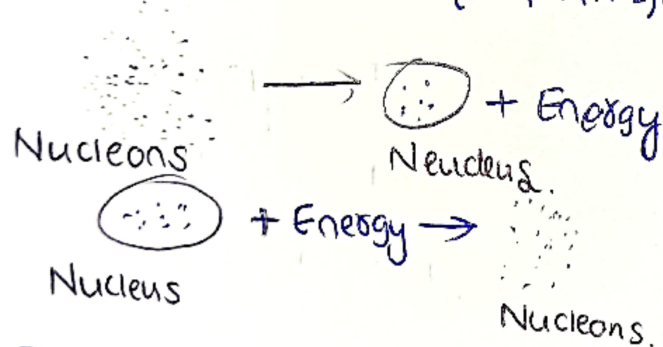
Nucleus $\begin{matrix} A \\ x \\ Z \end{matrix}$

Mass of Nucleus x is M_x

Mass of Z Protons is $z(m_p)$

Mass of $(A-Z)$ Neutrons is $(A-Z)(m_n)$

Mass of All Nucleons = $\{Zm_p + (A-Z)m_n\}$



- Binding Energy is the Energy released during the Formation of Nucleus from its constituent Nucleons ...
- Binding Energy is also Energy required to break the Nucleus into its constituent Nucleons.

$RHS > LHS$: Energy released

$LHS > RHS$: Energy Absorbed.

Nuclear Physics.

- Nuclear reactions/Transformations.
- Behaviour of Atomic Electrons not behaviour of Nuclei...

Nuclear Characteristics:

- Refer Module

Mass Defect (Δm)

- Mass of Nucleus is always less than the Mass of its Constituent Nucleons.

$$\Delta m = \{Z m_p + (A-Z) m_n\} - \{M_x\}$$

m_p : mass Proton

m_n : mass Neutron

m_x : mass Atom

Binding Energy:

$$BE = (\Delta m) c^2$$

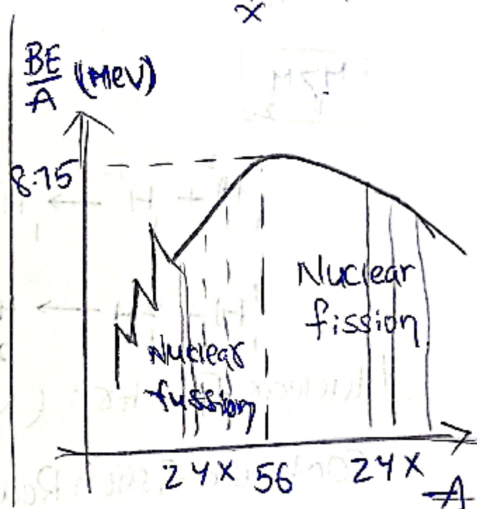
$$1 \text{ Amu} = 1.67 \times 10^{-27} \text{ Kg.}$$

$$E = (\Delta m) c^2$$

$$E = (1.67 \times 10^{-27}) (3 \times 10^8)^2 \text{ Joules}$$

$$E = \frac{(1.67 \times 10^{-27}) (9 \times 10^{16})}{1.6 \times 10^{-19} \times 10^6} \text{ MeV}$$

$$E = 931.5 \text{ MeV} = 1 \text{ u} = 1 \text{ amu}$$



Binding Energy per Nucleon: $\left(\frac{BE}{A}\right)$

Example: $^{88}_{20}X$ has a Total BE of 900 MeV, Then $\left(\frac{BE}{A}\right)_X = \frac{900}{88}$

$$\left(\frac{BE}{A}\right)_X = 4.5$$

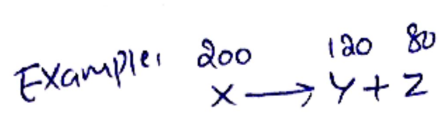
NOTE:

(i) Higher Value of $\left(\frac{BE}{A}\right)$ implies Greater Stability **

Of the Nucleus.

(ii) In Any Nuclear Rxn

$$\begin{array}{|c|c|} \hline X \rightarrow Y + Z & (BE)_{LHS} < (BE)_{RHS} \\ \hline X + Y \rightarrow Z & (M)_{LHS} > (M)_{RHS} \\ \hline \end{array}$$



$$\left(\frac{BE}{A}\right)_X = 6 \text{ MeV}$$

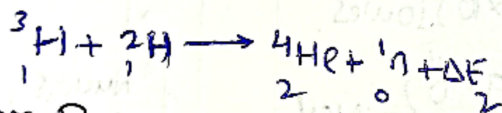
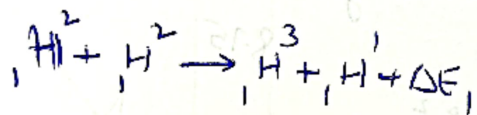
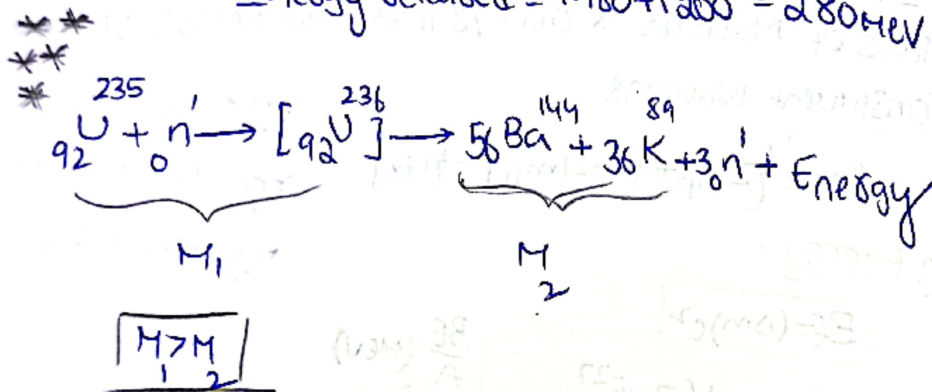
$$\left(\frac{BE}{A}\right)_Y = 7 \text{ MeV}$$

$$\left(\frac{BE}{A}\right)_Z = 8 \text{ MeV}$$

$$(BE)_X = 1200 \text{ MeV}$$

$$(BE)_{Y+Z} = 120(7) + 80(8) = 1480 \text{ MeV}$$

$$\text{Energy released} = 1480 + 1200 = 280 \text{ MeV}$$



Nuclear fusion...

Nuclear Reactor: (Nuclear controlled fission reaction)

Controlled fission reaction

1. Fuel Rods: $^{235}_{92}\text{U}$ rods
2. Moderator: To slow down the fast moving neutrons.
Heavy water, Graphite
3. Control Rod: To absorb the excess neutrons & thereby the reactions.
Boron, Cadmium rods...
4. Coolant: To remove the heat produced in the reactor...

